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SUBJECT CODE NO: - RNP-8009
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A & B) CSE / AI & ML / EEE (NEP) Sem - I
Examination June/July 2025
Basics of Electrical Engineering

[Time: 3:00 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

SECTION A**20 Marks****Q1 Solve any 10 from the following questions.**

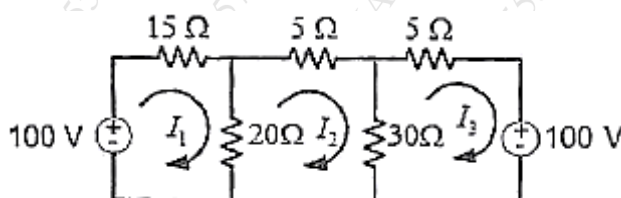
1. The algebraic sum of voltages around any closed path in a network is equal to _____
 A. ZERO B. Infinite C. low D. high
2. The resistance of a conductor is proportional to its _____
 A. Length B. Area C. current D. power
3. Line voltage in star connected three phase _____ times phase voltage.
 A. 3 time B. $\sqrt{3}$ C. Equal D. 2 Times
4. Line current in delta connected three phase _____ times line current.
 A. 3 time B. $\sqrt{3}$ C. Equal D. 2 Times
5. The value of form factor of sinusoidal waveform is _____
 A. 1.11 B. 1.21 C. 0.70 D. 0.63
6. RMS value of sinusoidal wave form is _____ Maximum value.
 A. 1.11 B. 1.21 C. 0.70 D. 0.63
7. The conductor and flux having angle between them 90 then flux link
 A. Maximum B. Minimum C. Equal D. Zero
8. The ratio of maximum value to the r.m.s. value of an alternating quantity is known _____
 A. Zero B. Peak Factor C. Form Factor D. None
9. The flux variations being brought about by variations
 A. Density B. voltage C. current D. Power
10. An ideal voltage source is that which has an internal resistance
 A. Zero B. Low C. High D. Infinite

11. The phenomenon of lagging of flux density (B) behind the magnetising force (H) in a magnetic material
 A. Flux B. Current C. B-H Curve D. Voltage
12. In Kirchhoff's first law $\sum i = 0$ at the junction is based on the conservation of
 A. Charge B. Current C. Energy D. Density
13. The resistance of a material is most commonly determined by four factors-length, cross-sectional area, type of material and _____
 A. Current B. Voltage C. Power D. Temperature

SECTION B**40 Marks**

- Q2** a) Prove the Series and parallel combination of resistor with circuit diagram. **05**
 b) Expression for value of α at different temperature in detail. **05**

- Q3** a) Find loop Current I_1 , I_2 & I_3 By Mesh Analysis. **05**



- b) Explain Superposition theorem with an example. **05**
- Q4** a) Comparison between Magnetic and Electric Circuits. **05**
 b) A solenoid have 800 turns supply with 30 volt of resistance is 6 ohm, length of solenoid is 16 cm with area is 6 sq.cm, relative permeability is 800 find
 1) MMF 2) Magnetic field intensity 3) Flux density 4) Reluctance **05**
- Q5** a) Prove the Coefficient of Coupling with equation. **05**
 b) Prove the Capacitors In Series with detail. **05**
- Q6** a) Explain R-L Series A.C. Circuit with phasor diagram **05**
 b) Explain R-C Series A.C. Circuit. **05**
- Q7** a) Explain the relation between Star connected phase voltage and line voltage in detail **05**
 b) Explain the relation between Delta connected phase current and line current in detail **05**